

1. Executive Summary

The insurance industry is at a pivotal inflection point. Decades of manual workflows, fragmented legacy systems, and paper-based processes are colliding with an exponential AI capability.

The result: insurers that move now will define the competitive landscape for the next decade. Those who wait will find themselves structurally unable to compete.

The problem:

- 70% of insurer IT budgets are consumed by maintaining legacy systems.
- 1 in 10 insurers have less than 50% legacy infrastructure.
- P&C insurance fraud costs \$90-122 billion annually.
- Claims processing averages 7-30 days; customer expectations have shifted to real-time.

The AI opportunity:

- AI in the insurance market is valued at \$10.36 billion in 2025, projected to reach \$154.39 billion by 2034 at 35.7%.
- Full AI adoption among insurers jumped from 8% to 34% year-over-year.
- AI-powered claims automation reduces processing time by up to 70%, saving an estimated \$6.5 billion annually.
- Machine learning in underwriting improves accuracy by 54%, processing time from weeks to hours.

In short: AI delivers three simultaneous levers: cost reduction (30-40% operational savings), revenue growth (better pricing accuracy, cross-sell retention), and risk mitigation (fraud detection, compliance automation). No single prior technology investment has offered all three at this scale and speed.



Where to Invest: Priority Areas

Function	AI Application	Typical ROI Signal
Claims	Automated FNOL, damage assessment, STP	70% faster settlement rate, 30% cost reduction
Underwriting	ML risk models, real-time pricing	54% accuracy improvement, 90% faster quotes
Fraud detection	Behavioral analytics, NLP anomaly detection	20-40% fraud reduction, \$7.5 billion saved globally
Customer service	Conversational AI, copilots	42% of queries handled, 15-20% CSAT uplift
Back office	RPA, document processing, compliance	80% manual effort reduction in data entry

2. Industry Context: Why AI is Reshaping Insurance

Current challenges

Insurance remains one of the most data-intensive industries, yet one of the least digitalized in its core operations. The gap between what is technologically possible and what is operationally practiced is enormous, and that gap is the primary driver of AI adoption urgency.

- 70% of insurer IT budgets are spent on maintaining legacy systems rather than innovation.
- Manual data entry dominates back-office operations where a single file touches 15 systems.
- IT's share of operating costs rose from 17% to 24% for P&C carriers between 2012 and 2017, continuing upward.
- 74% of insurance companies still rely on outdated technology.

Growing customer expectations

- Customers now compare insurer digital experience to Amazon, Uber, and fintech apps.
- Digital-first competitors set new benchmarks—claims in seconds, policies in minutes.
- Embedded insurance is projected to grow gross premiums 6x to \$722 billion by 2030.

Fraud escalation

- Global insurance fraud losses exceed \$80 billion annually
- US P&C fraud alone was \$90-122 billion annually
- Insurance fraud rose 19% year-over-year in 2024



Market Drivers

Driver	Impact on AI Adoption
Exponential data growth	IoT devices, telematics, and wearables are generating real-time risk signals
InsurTech competition	Digital-native players operating at 10-15x lower expense ratios, forcing incumbents to match speed and personalization
Regulatory and compliance pressure	HIPAA non-compliance averaged \$1.45M per incident in 2024; AI-driven compliance automation is becoming table stakes
Cost pressure	Combined ratios under strain, insurers are not earning the cost of capital without operational transformation
Investor pressure	InsurTech investment and M&A activity signaling urgency, incumbents acquiring or partnering to accelerate capability



The Dilemma

Despite clear evidence of AI's value, many insurers remain hesitant. This is not irrational; the barriers are real.

Business Case Uncertainty

- ROI on AI is non-linear—early phases show cost before benefit
- Multi-year transformation programmes compete with quarterly earnings expectations
- Siloed P&L structures make it difficult to attribute AI value across functions
- 52% of insurers cite skills and resource constraints as the primary AI implementation barrier

Technology Challenges

- Legacy core systems not designed for API integration with AI tools
- Data quality and consistency issues as AI models require clean, structured, consistent data
- Multiple overlapping systems of record create data governance complexity
- Vendor lock-in risk and difficulty in selecting between build vs. buy vs. partner models

Security and Regulatory Concerns

- AI model explainability requirements, where regulators increasingly demand auditable decision trails
- EU AI Act, NIST AI RMF, and ISO/IEC 42001 all require embedded governance in AI workflows
- Model bias concerns in underwriting and claims—risk of discriminatory outcomes
- Adversarial AI threats: fraudsters using AI faster than insurers can defend against it
- 36% of insurers cite regulatory hurdles as a top AI adoption barrier

The Overall Dilemma: The cost of inaction is higher than the cost of investment— but the path forward requires managing real technology, organisational, and regulatory risk simultaneously.